

□ Exploratory Data Analysis — Sample Superstore

□ About the Dataset and Analysis

About the Dataset

The **Sample Superstore Dataset** is a retail sales dataset that contains detailed information about customer orders, products, sales transactions, shipping methods, geographical locations, discounts, and profits. It is commonly used for learning and demonstrating **Exploratory Data Analysis (EDA), data visualization, and business analytics**.

The dataset originally contains **9,994 records and 13 columns**. The columns provide information about shipping, customers, locations, products, sales, quantity, discounts, and profit. Important categorical columns include **Ship Mode, Segment, Country, City, State, Region, Category, and Sub-Category**. The numerical columns include **Postal Code, Sales, Quantity, Discount, and Profit**.

This dataset is useful for understanding how different business factors influence sales and profitability. It allows analysts to investigate questions such as which categories generate the highest revenue, which regions perform best, how discounts affect profit, and which products or sub-categories are generating losses.

About the Analysis

The main purpose of this project is to perform a complete **Exploratory Data Analysis (EDA)** before drawing business conclusions. The analysis begins with loading and inspecting the dataset using **Python, Pandas, NumPy, Matplotlib, and Seaborn**.

The initial analysis checks the dataset shape, column names, data types, missing values, and duplicate records. The dataset contains **no missing values**, while **17 duplicate records** were identified. After removing these duplicates, the final dataset contains **9,977 unique records**.

Next, descriptive statistics are calculated using `describe()` to understand the distribution of numerical variables such as Sales, Quantity, Discount, and Profit. The analysis shows that sales values are highly right-skewed because a small number of transactions have very high sales values.

Several visualizations are created to understand the data more clearly. **Histograms** are used to study Sales, Profit, and Quantity distributions. **Boxplots** are used to identify potential outliers. A **correlation heatmap** is created to understand relationships between numerical variables. **Bar charts** are used to compare sales and profit across categories, sub-categories, regions, and customer segments. **Scatter plots** are used to investigate relationships such as Sales vs Profit and Discount vs Profit.

The analysis shows that the business generated approximately **\$2.30 million in sales and \$286,000 in profit**, with an overall profit margin of approximately **12.5%**. Technology is the

strongest category in terms of both sales and profit, while Furniture has a much lower profit margin. The **Tables** sub-category is particularly concerning because it generates a significant overall loss.

Another important finding is the relationship between discount and profitability. Higher discounts are generally associated with lower profits, and very high discount levels can result in negative profitability. The analysis also shows a moderate positive relationship between Sales and Profit, but high sales do not always guarantee high profit.

Overall, this EDA project demonstrates how statistical analysis and visualization can transform raw retail data into meaningful **business insights and recommendations**. It provides a strong foundation for further analysis, reporting, and data-driven decision-making.

```
import pandas as pd          # Data manipulation and analysis
import numpy as np           # Numerical calculations

import matplotlib.pyplot as plt # Data visualization
import seaborn as sns         # Statistical data visualization

from scipy import stats       # Statistical analysis

import warnings               # Handle warning messages
warnings.filterwarnings("ignore")
sns.set_theme(style="whitegrid")

df = pd.read_csv("SampleSuperstore.csv")
df.head(10) # Top 10 rowdf.
```

	Ship Mode	Segment	Country	City
State \				
0	Second Class	Consumer	United States	Henderson
Kentucky				
1	Second Class	Consumer	United States	Henderson
Kentucky				
2	Second Class	Corporate	United States	Los Angeles
California				
3	Standard Class	Consumer	United States	Fort Lauderdale
Florida				
4	Standard Class	Consumer	United States	Fort Lauderdale
Florida				
5	Standard Class	Consumer	United States	Los Angeles
California				
6	Standard Class	Consumer	United States	Los Angeles
California				
7	Standard Class	Consumer	United States	Los Angeles
California				
8	Standard Class	Consumer	United States	Los Angeles
California				
9	Standard Class	Consumer	United States	Los Angeles
California				

Postal Code	Region	Category	Sub-Category	Sales
Quantity \				
0 42420	South	Furniture	Bookcases	261.9600
2				
1 42420	South	Furniture	Chairs	731.9400
3				
2 90036	West	Office Supplies	Labels	14.6200
2				
3 33311	South	Furniture	Tables	957.5775
5				
4 33311	South	Office Supplies	Storage	22.3680
2				
5 90032	West	Furniture	Furnishings	48.8600
7				
6 90032	West	Office Supplies	Art	7.2800
4				
7 90032	West	Technology	Phones	907.1520
6				
8 90032	West	Office Supplies	Binders	18.5040
3				
9 90032	West	Office Supplies	Appliances	114.9000
5				

Discount	Profit
0 0.00	41.9136
1 0.00	219.5820
2 0.00	6.8714
3 0.45	-383.0310
4 0.20	2.5164
5 0.00	14.1694
6 0.00	1.9656
7 0.20	90.7152
8 0.20	5.7825
9 0.00	34.4700

df.tail(10) # Last 10 row

Ship Mode	Segment	Country	City
State \			
9984 Standard Class	Consumer	United States	Long Beach New York
9985 Standard Class	Consumer	United States	Long Beach New York
9986 Standard Class	Consumer	United States	Los Angeles California
9987 Standard Class	Corporate	United States	Athens Georgia
9988 Standard Class	Corporate	United States	Athens Georgia

9989	Second Class	Consumer	United States	Miami Florida
9990	Standard Class	Consumer	United States	Costa Mesa California
9991	Standard Class	Consumer	United States	Costa Mesa California
9992	Standard Class	Consumer	United States	Costa Mesa California
9993	Second Class	Consumer	United States	Westminster California

	Postal Code	Region	Category	Sub-Category	Sales Quantity \
9984	11561	East	Office Supplies	Labels	31.500
9985	11561	East	Office Supplies	Supplies	55.600
9986	90008	West	Technology	Accessories	36.240
9987	30605	South	Technology	Accessories	79.990
9988	30605	South	Technology	Phones	206.100
9989	33180	South	Furniture	Furnishings	25.248
9990	92627	West	Furniture	Furnishings	91.960
9991	92627	West	Technology	Phones	258.576
9992	92627	West	Office Supplies	Paper	29.600
9993	92683	West	Office Supplies	Appliances	243.160

	Discount	Profit
9984	0.0	15.1200
9985	0.0	16.1240
9986	0.0	15.2208
9987	0.0	28.7964
9988	0.0	55.6470
9989	0.2	4.1028
9990	0.0	15.6332
9991	0.2	19.3932
9992	0.0	13.3200
9993	0.0	72.9480

df.size # Size of Dataset

129922

```
df.columns # All columns Name
Index(['Ship Mode', 'Segment', 'Country', 'City', 'State', 'Postal
Code',
      'Region', 'Category', 'Sub-Category', 'Sales', 'Quantity',
      'Discount',
      'Profit'],
      dtype='object')

df.shape

(9994, 13)
```

Data types

```
df.dtypes # Data types

Ship Mode      object
Segment        object
Country        object
City           object
State          object
Postal Code    int64
Region         object
Category       object
Sub-Category   object
Sales          float64
Quantity       int64
Discount       float64
Profit         float64
dtype: object

The history saving thread hit an unexpected error
(OperationalError('attempt to write a readonly database')).History
will not be written to the database.
```

Information

```
df.info() # Information

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Ship Mode      9994 non-null  object
 1   Segment       9994 non-null  object
```

```

2   Country      9994 non-null    object
3   City         9994 non-null    object
4   State        9994 non-null    object
5   Postal Code  9994 non-null    int64
6   Region       9994 non-null    object
7   Category     9994 non-null    object
8   Sub-Category 9994 non-null    object
9   Sales        9994 non-null    float64
10  Quantity     9994 non-null    int64
11  Discount     9994 non-null    float64
12  Profit       9994 non-null    float64
dtypes: float64(3), int64(2), object(8)
memory usage: 1015.1+ KB

```

Check the number of missing (NaN) values in each column

```
df.isnull().sum()## Check the number of missing (NaN) values in each column
```

```

Ship Mode      0
Segment        0
Country        0
City           0
State          0
Postal Code    0
Region         0
Category       0
Sub-Category   0
Sales          0
Quantity       0
Discount       0
Profit         0
dtype: int64

```

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
```

```
missing_percentage
```

```

Ship Mode      0.0
Segment        0.0
Country        0.0
City           0.0
State          0.0
Postal Code    0.0
Region         0.0
Category       0.0

```

```
Sub-Category    0.0
Sales           0.0
Quantity        0.0
Discount        0.0
Profit          0.0
dtype: float64
```

Count the total number of duplicate

```
df.duplicated().sum() # Count the total number of duplicate
np.int64(17)
df_clean = df.copy()
df_clean = df_clean.drop_duplicates()
df_clean.shape
(9977, 13)

# Find duplicate rows in the DataFrame
duplicate_rows = df[df.duplicated()] # df.duplicated() returns True
for duplicate rows (after the first occurrence)

# Print message for clarity
print("\nDuplicate Rows:")

# Display the duplicate rows
print(duplicate_rows)

Duplicate Rows:
Empty DataFrame
Columns: [Ship Mode, Segment, Country, City, State, Postal Code,
Region, Category, Sub-Category, Sales, Quantity, Discount, Profit]
Index: []
```

Remove Duplicate

```
# Drop duplicate rows based on all columns
df = df.drop_duplicates()

# Verify
print("After dropping duplicates, shape:", df.shape)

After dropping duplicates, shape: (9977, 13)
```

Statistical Summary

```
df_clean.describe()
```

	Postal Code	Sales	Quantity	Discount
Profit				
count	9977.000000	9977.000000	9977.000000	9977.000000
mean	55154.964117	230.148902	3.790719	0.156278
std	32058.266816	623.721409	2.226657	0.206455
min	1040.000000	0.444000	1.000000	0.000000
25%	23223.000000	17.300000	2.000000	0.000000
50%	55901.000000	54.816000	3.000000	0.200000
75%	90008.000000	209.970000	5.000000	0.200000
max	99301.000000	22638.480000	14.000000	0.800000

Numerical Summary

```
df_clean[["Sales", "Quantity", "Discount", "Profit"]].describe()
```

	Sales	Quantity	Discount	Profit
count	9977.000000	9977.000000	9977.000000	9977.000000
mean	230.148902	3.790719	0.156278	28.69013
std	623.721409	2.226657	0.206455	234.45784
min	0.444000	1.000000	0.000000	-6599.97800
25%	17.300000	2.000000	0.000000	1.72620
50%	54.816000	3.000000	0.200000	8.67100
75%	209.970000	5.000000	0.200000	29.37200
max	22638.480000	14.000000	0.800000	8399.97600

Calculate Total Sales

```
df_clean["Sales"].sum()
```

```
np.float64(2296195.5903)
```

Calculate Total Profit

```
df_clean["Profit"].sum()  
np.float64(286241.4226)
```

Calculate Total Quantity

```
df_clean["Quantity"].sum()  
np.int64(37820)
```

Calculate Overall Profit Margin

```
total_sales = df_clean["Sales"].sum()  
total_profit = df_clean["Profit"].sum()  
  
profit_margin = (total_profit / total_sales) * 100  
  
profit_margin  
np.float64(12.465898976950927)
```

Analyze Categorical Variables

```
df_clean["Ship Mode"].value_counts()  
  
Ship Mode  
Standard Class      5955  
Second Class        1943  
First Class          1537  
Same Day             542  
Name: count, dtype: int64
```

Standard Class is clearly the dominant shipping method, indicating that customers frequently prefer the lower-cost standard delivery option.

Segment Analysis

```
df_clean["Segment"].value_counts()  
  
Segment  
Consumer      5183  
Corporate     3015
```

```
Home Office      1779
Name: count, dtype: int64
```

Consumer is the largest customer segment.

Region Analysis

```
df_clean["Region"].value_counts()
```

```
Region
West      3193
East      2845
Central   2319
South     1620
Name: count, dtype: int64
```

The West region has the largest number of records.

Category Analysis

```
df_clean["Category"].value_counts()
```

```
Category
Office Supplies    6012
Furniture           2118
Technology          1847
Name: count, dtype: int64
```

Office Supplies has the highest number of transactions.

Sub-Category Analysis

```
df_clean["Sub-Category"].value_counts()
```

```
Sub-Category
Binders      1522
Paper        1359
Furnishings   956
Phones       889
Storage       846
Art           795
Accessories   775
Chairs        615
Appliances    466
Labels        363
```

```
Tables      319
Envelopes   254
Bookcases   228
Fasteners   217
Supplies    190
Machines    115
Copiers      68
Name: count, dtype: int64
```

-----Visualization 1 — Sales Distribution-----

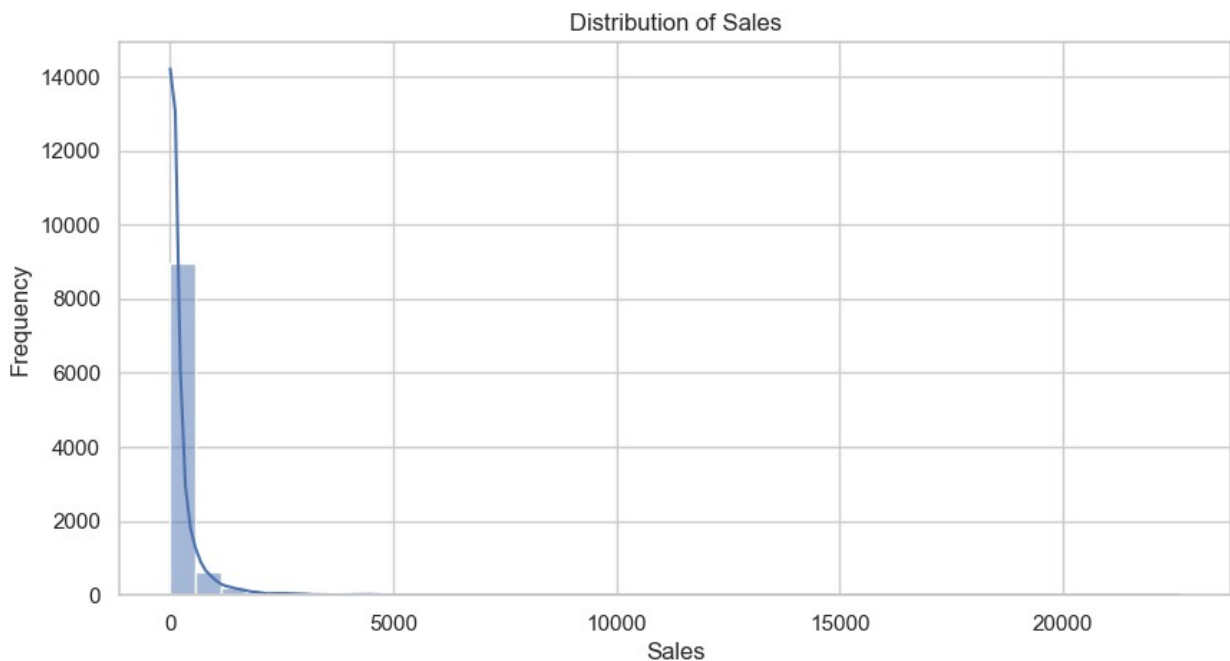
Visualization 1 — Sales Distribution

```
plt.figure(figsize=(10, 5))

sns.histplot(df_clean["Sales"], bins=40, kde=True)

plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

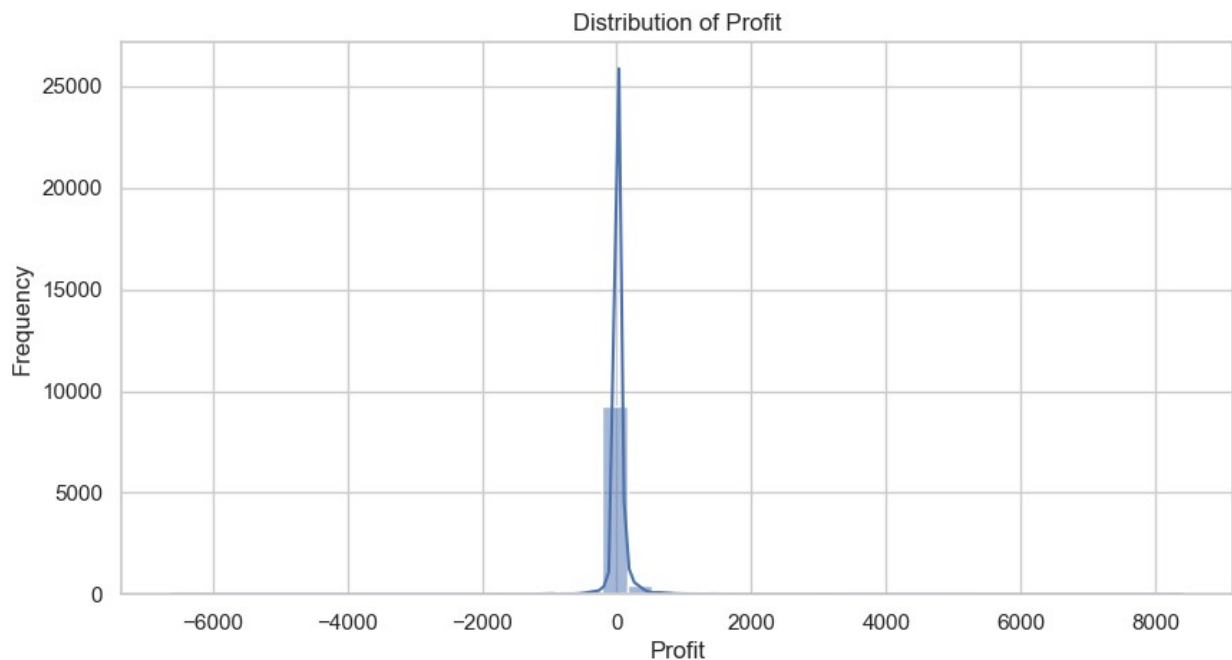
plt.show()
```



Sales are right-skewed. Most transactions have relatively low sales values, while a small number of transactions have extremely high sales.

Visualization 2 — Profit Distribution

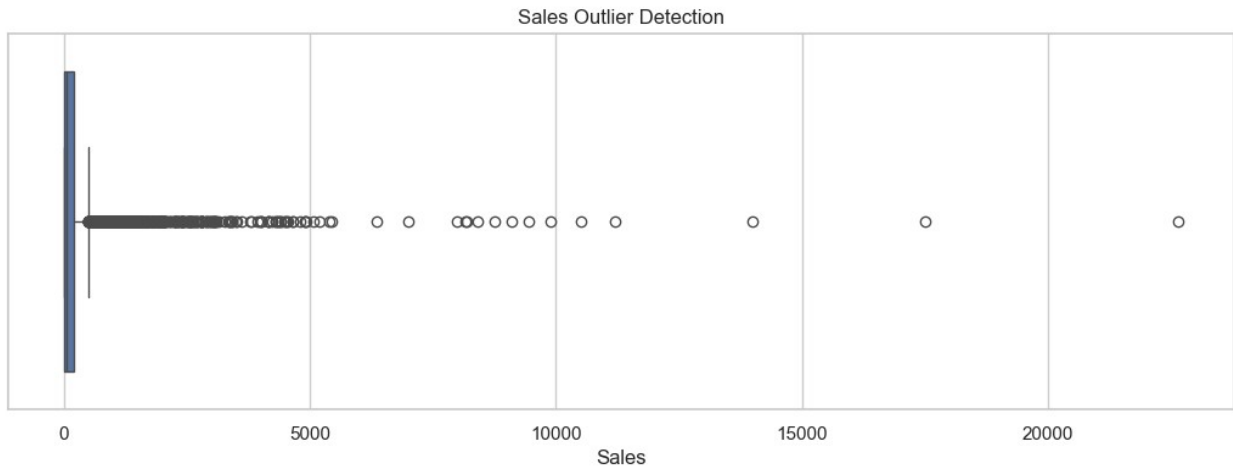
```
plt.figure(figsize=(10, 5))  
  
sns.histplot(df_clean["Profit"], bins=40, kde=True)  
  
plt.title("Distribution of Profit")  
plt.xlabel("Profit")  
plt.ylabel("Frequency")  
  
plt.show()
```



Most transactions generate relatively small profits. The dataset also contains negative-profit transactions, indicating that some sales resulted in losses.

Visualization 3 — Boxplot for Sales

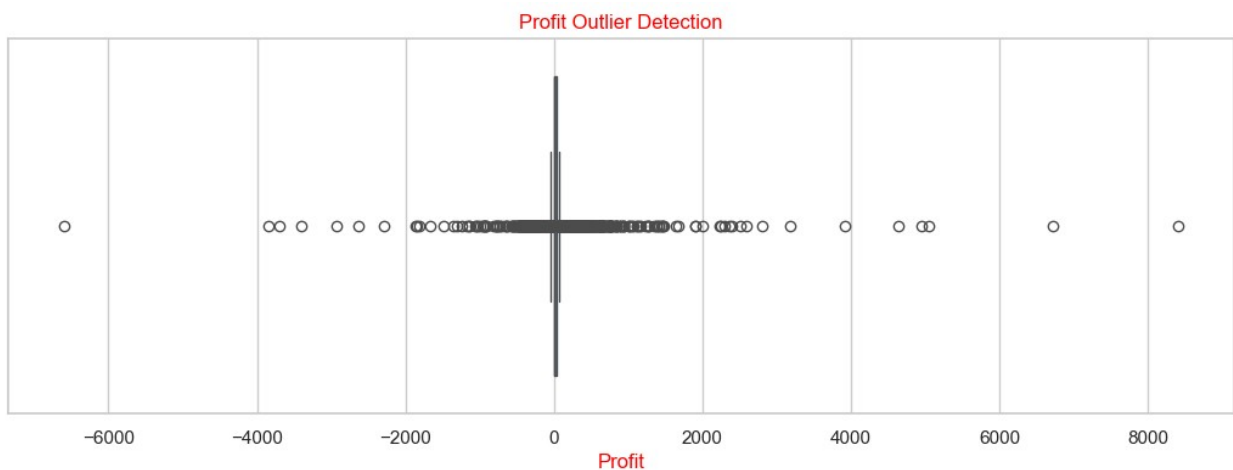
```
plt.figure(figsize=(13, 4))  
  
sns.boxplot(x=df_clean["Sales"])  
  
plt.title("Sales Outlier Detection")  
plt.xlabel("Sales")  
  
plt.show()
```



Several high-value sales transactions appear as potential outliers.

Visualization 4 — Boxplot for Profit

```
plt.figure(figsize=(13, 4))
sns.boxplot(x=df_clean["Profit"])
plt.title("Profit Outlier Detection", color="red")
plt.xlabel("Profit", color="red")
plt.show()
```



Profit contains several extreme positive and negative observations. These should be investigated rather than automatically removed.

Statistical Outlier Detection — IQR

```
Q1 = df_clean["Sales"].quantile(0.25)
Q3 = df_clean["Sales"].quantile(0.75)
```

```

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)

Lower Bound: -271.705
Upper Bound: 498.975

sales_outliers = df_clean[
    (df_clean["Sales"] < lower_bound) |
    (df_clean["Sales"] > upper_bound)
]

sales_outliers.shape

(1167, 13)

```

Check Negative Profit

```

negative_profit = df_clean[df_clean["Profit"] < 0]

negative_profit.shape

(1869, 13)

```

A significant number of transactions generate negative profit. This is an important business issue and deserves further analysis.

Correlation Matrix

```

correlation = df_clean[
    ["Sales", "Quantity", "Discount", "Profit"]
].corr()

correlation

```

	Sales	Quantity	Discount	Profit
Sales	1.000000	0.200722	-0.028311	0.479067
Quantity	0.200722	1.000000	0.008678	0.066211
Discount	-0.028311	0.008678	1.000000	-0.219662
Profit	0.479067	0.066211	-0.219662	1.000000

```

correlation = df_clean[
    ["Sales", "Quantity", "Discount", "Profit"]
].corr()

correlation

```

	Sales	Quantity	Discount	Profit
Sales	1.000000	0.200722	-0.028311	0.479067
Quantity	0.200722	1.000000	0.008678	0.066211
Discount	-0.028311	0.008678	1.000000	-0.219662
Profit	0.479067	0.066211	-0.219662	1.000000

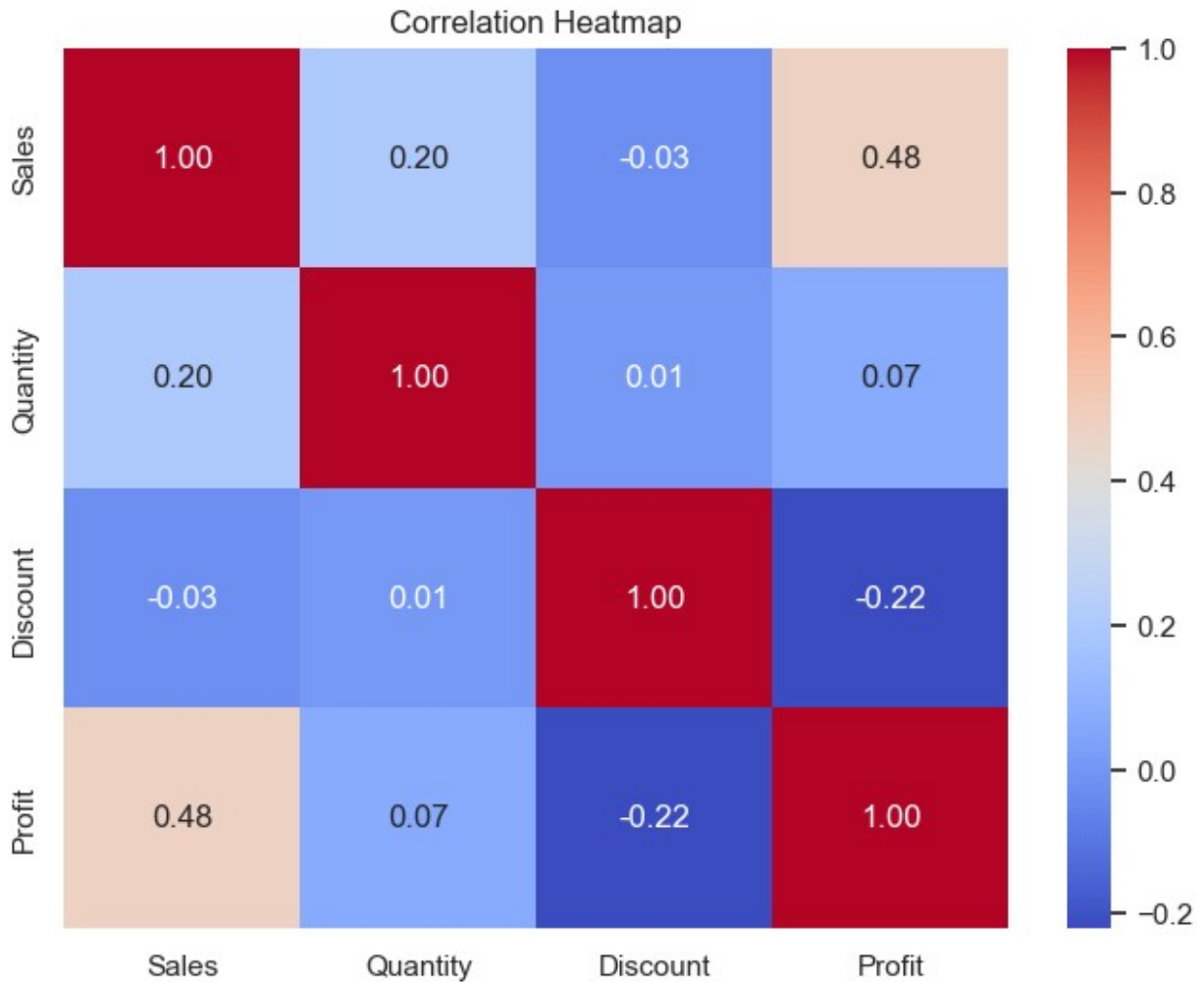
Correlation Heatmap

```
plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()
```



Category-wise Sales

```
category_sales = (  
    df_clean.groupby("Category")["Sales"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

category_sales

```
Category  
Technology      836154.0330  
Furniture       741306.3133  
Office Supplies 718735.2440  
Name: Sales, dtype: float64
```

Technology generates the highest total sales, despite Office Supplies having the largest number of transactions.

Category-wise Profit

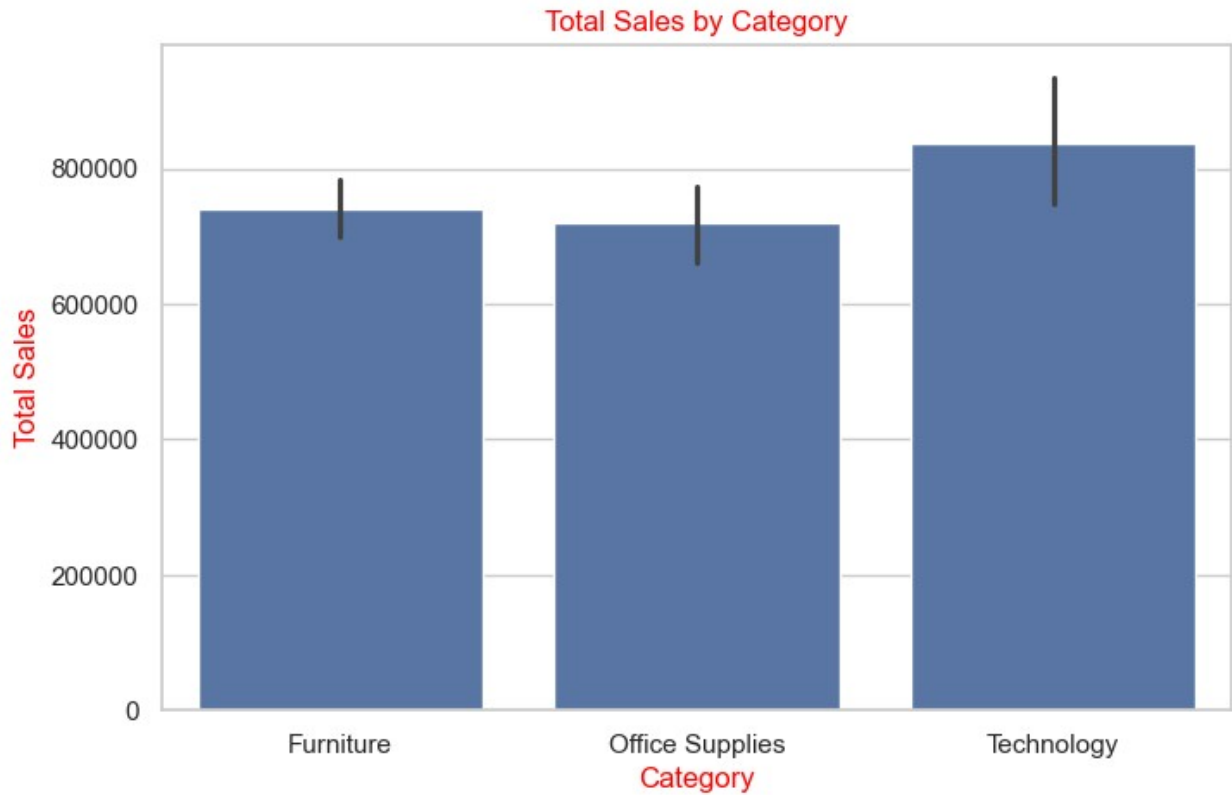
```
category_profit = (  
    df_clean.groupby("Category")["Profit"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

category_profit

```
Category  
Technology      145454.9481  
Office Supplies  122364.6608  
Furniture       18421.8137  
Name: Profit, dtype: float64
```

Category Sales Visualization

```
plt.figure(figsize=(8, 5))  
  
sns.barplot(  
    data=df_clean,  
    x="Category",  
    y="Sales",  
    estimator="sum"  
)  
  
plt.title("Total Sales by Category", color="Red")  
plt.xlabel("Category", color="Red")  
plt.ylabel("Total Sales", color="Red")  
  
plt.show()
```



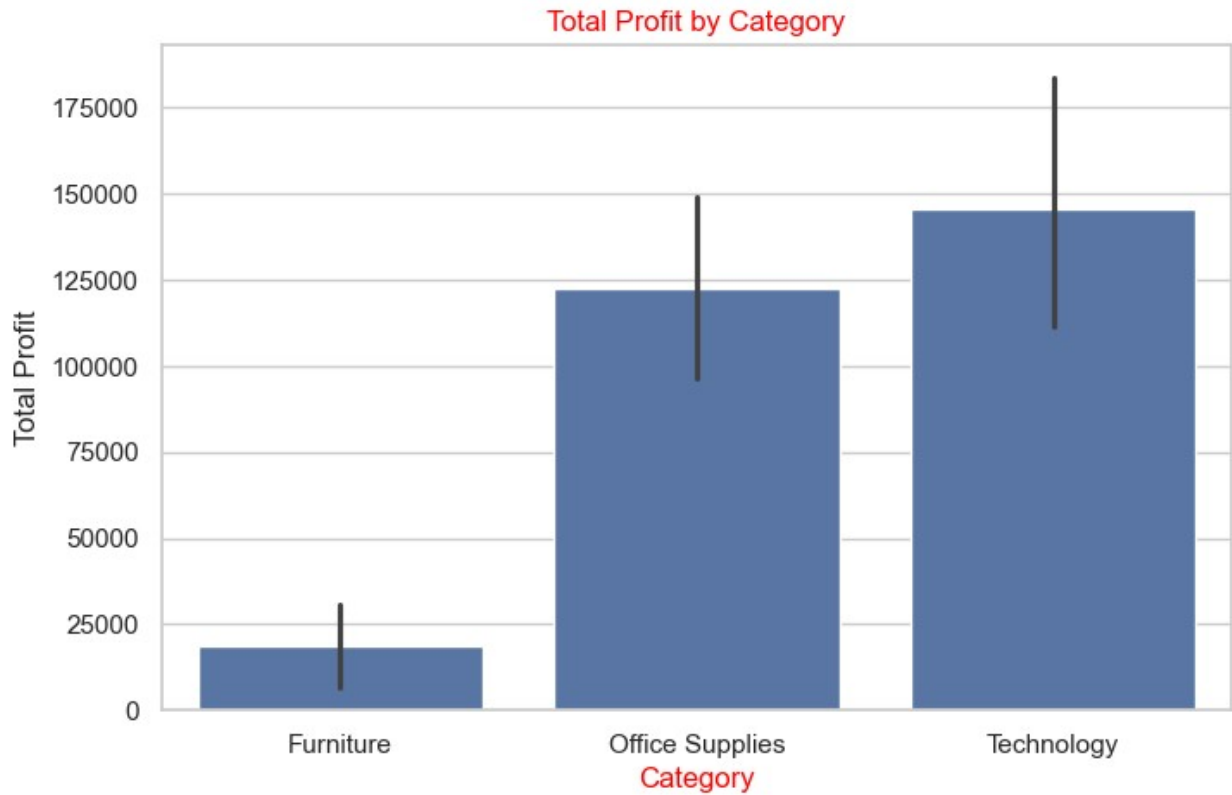
Category Profit Visualization

```
plt.figure(figsize=(8, 5))

sns.barplot(
    data=df_clean,
    x="Category",
    y="Profit",
    estimator="sum"
)

plt.title("Total Profit by Category",color="Red")
plt.xlabel("Category",color="Red")
plt.ylabel("Total Profit")

plt.show()
```



Sub-Category Profitability

```
subcategory_profit = (  
    df_clean.groupby("Sub-Category")["Profit"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

subcategory_profit

Sub-Category

Copiers	55617.8249
Phones	44515.7306
Accessories	41936.6357
Paper	33944.2395
Binders	30228.0003
Chairs	26567.1278
Storage	21278.8264
Appliances	18138.0054
Furnishings	13052.7230
Envelopes	6964.1767
Art	6524.6118
Labels	5526.3820
Machines	3384.7569

```
Fasteners      949.5182
Supplies      -1189.0995
Bookcases     -3472.5560
Tables        -17725.4811
Name: Profit, dtype: float64
```

Sub-Category Profit Visualization

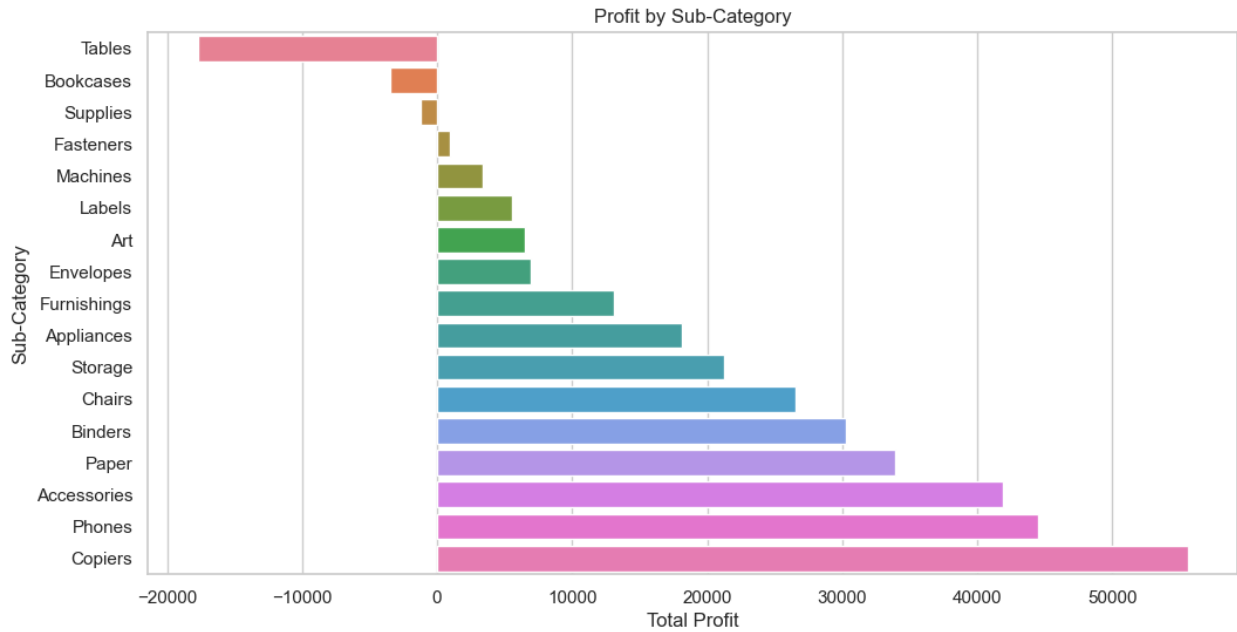
```
plt.figure(figsize=(12, 6))

profit_data = (
    df_clean.groupby("Sub-Category")["Profit"]
    .sum()
    .sort_values()
)

sns.barplot(
    x=profit_data.values,
    y=profit_data.index,
    palette="husl",
    hue=profit_data.index,
    legend=False
)

plt.title("Profit by Sub-Category")
plt.xlabel("Total Profit")
plt.ylabel("Sub-Category")

plt.show()
```



Sales vs Profit Analysis

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df_clean,
    x="Sales",
    y="Profit",
    alpha=0.5
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()
```



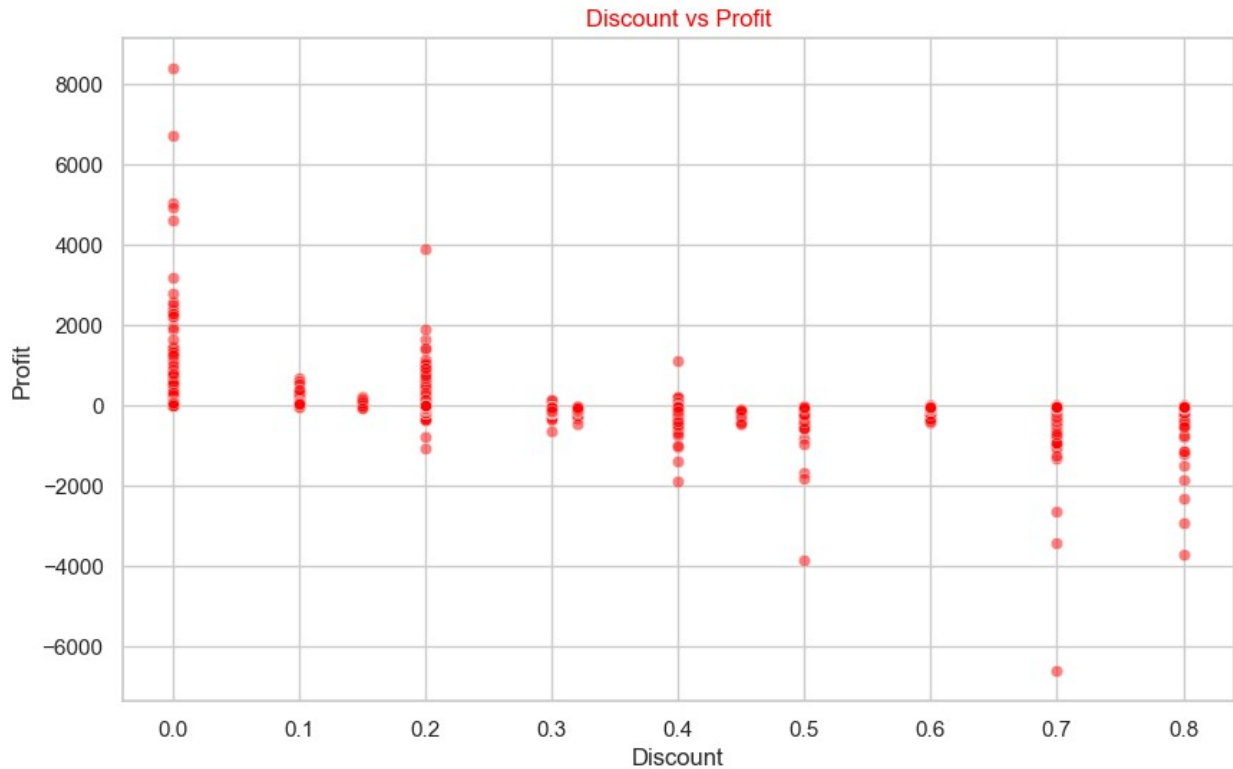
Discount vs Profit

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df_clean,
    x="Discount",
    y="Profit",
    alpha=0.5,
    color="red"
)

plt.title("Discount vs Profit", color="red")
plt.xlabel("Discount")
plt.ylabel("Profit")

plt.show()
```



Discount-wise Profit Analysis

```
discount_analysis = (
    df_clean.groupby("Discount")
    .agg(
        Sales=("Sales", "sum"),
        Profit=("Profit", "sum"),
        Transactions=("Profit", "size")
    )
)
```

```
discount_analysis["Profit_Margin_%"] = (
    discount_analysis["Profit"] /
    discount_analysis["Sales"] * 100
)
```

discount_analysis

	Sales	Profit	Transactions	Profit_Margin_%
Discount				
0.00	1.087278e+06	320844.4073	4787	29.508970
0.10	5.436935e+04	9029.1770	94	16.607108
0.15	2.755852e+04	1418.9915	52	5.149012
0.20	7.645049e+05	90306.6070	3653	11.812429
0.30	1.029453e+05	-10357.2186	226	-10.060897
0.32	1.449346e+04	-2391.1377	27	-16.498047

0.40	1.164178e+05	-23057.0504	206	-19.805437
0.45	5.484974e+03	-2493.1111	11	-45.453472
0.50	5.891854e+04	-20506.4281	66	-34.804712
0.60	6.644700e+03	-5944.6552	138	-89.464614
0.70	4.062028e+04	-40075.3569	418	-98.658490
0.80	1.696019e+04	-30532.8022	299	-180.026277

Business observation:- Excessive discounting appears to be strongly associated with loss-making transactions.

Region-wise Sales

```
region_sales = (
    df_clean.groupby("Region")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

region_sales
```

Region	Sales
West	725255.6365
East	678435.1960
Central	500782.8528
South	391721.9050

Name: Sales, dtype: float64

Observation:- The West region generates the highest sales.

Region-wise Profit

```
region_profit = (
    df_clean.groupby("Region")["Profit"]
    .sum()
    .sort_values(ascending=False)
)

region_profit
```

Region	Profit
West	108329.8079
East	91506.3092
South	46749.4303
Central	39655.8752

Name: Profit, dtype: float64

Observation:- West is also the most profitable region. However, Central has a relatively weak profit margin compared with the other regions.

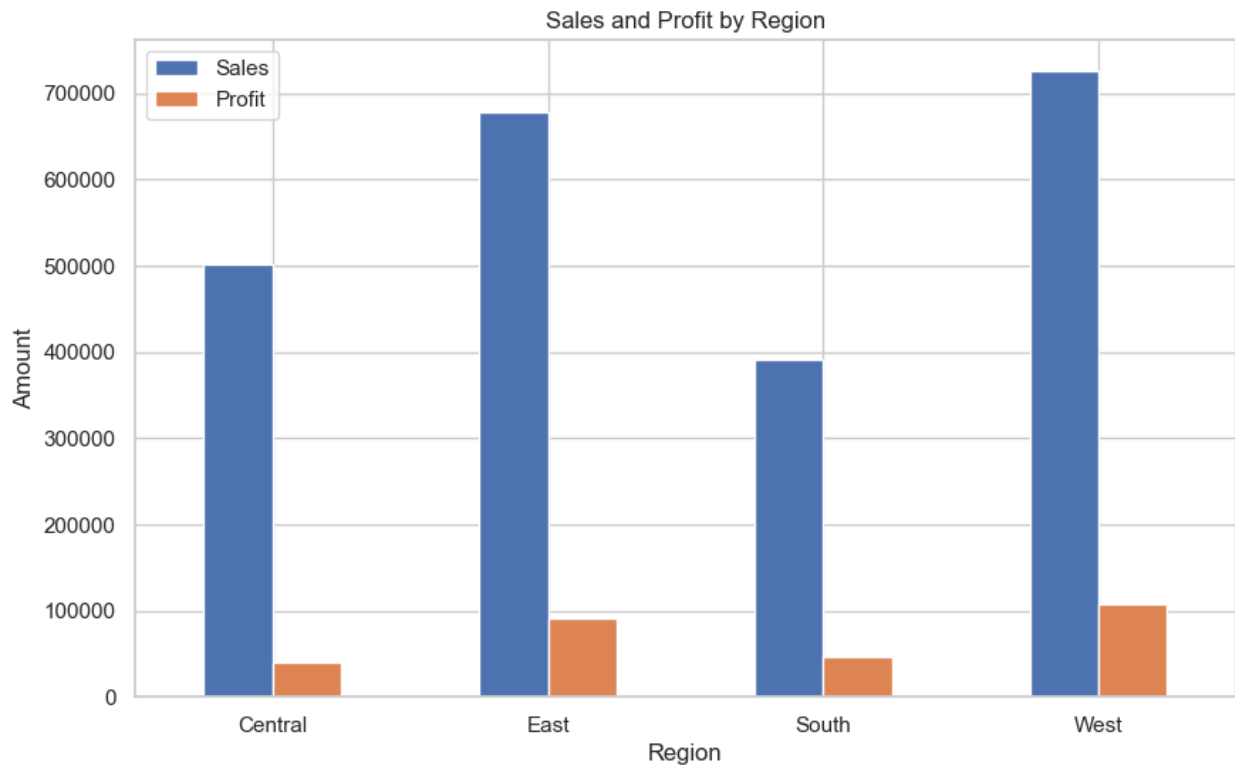
Region Sales & Profit Visualization

```
region_data = df_clean.groupby("Region")[["Sales", "Profit"]].sum()

region_data.plot(
    kind="bar",
    figsize=(10, 6)
)

plt.title("Sales and Profit by Region")
plt.xlabel("Region")
plt.ylabel("Amount")

plt.xticks(rotation=0)
plt.show()
```



Segment-wise Sales

```
segment_sales = (
    df_clean.groupby("Segment")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

segment_sales
```

```
Segment
Consumer      1.160833e+06
Corporate      7.060701e+05
Home Office    4.292927e+05
Name: Sales, dtype: float64
```

Observation:- The Consumer segment generates the highest sales.

Segment-wise Profit

```
segment_profit = (
    df_clean.groupby("Segment")["Profit"]
    .sum()
    .sort_values(ascending=False)
)
```

```
segment_profit
```

```
Segment
Consumer      134007.4413
Corporate       91954.9798
Home Office     60279.0015
Name: Profit, dtype: float64
```

Observation :- Consumer also generates the highest total profit. However, when comparing profit margins, Home Office has a stronger margin than Consumer.

Ship Mode Analysis

```
ship_analysis = (
    df_clean.groupby("Ship Mode")
    .agg(
        Sales=("Sales", "sum"),
        Profit=("Profit", "sum"),
        Quantity=("Quantity", "sum")
    )
)
```

```
ship_analysis
```

	Sales	Profit	Quantity
Ship Mode			
First Class	3.513805e+05	48953.6561	5690
Same Day	1.283217e+05	15871.8869	1956
Second Class	4.591770e+05	57446.6516	7418
Standard Class	1.357316e+06	163969.2280	22756

Top 10 Cities by Sales

```
top_cities = (
    df_clean.groupby("City")["Sales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_cities

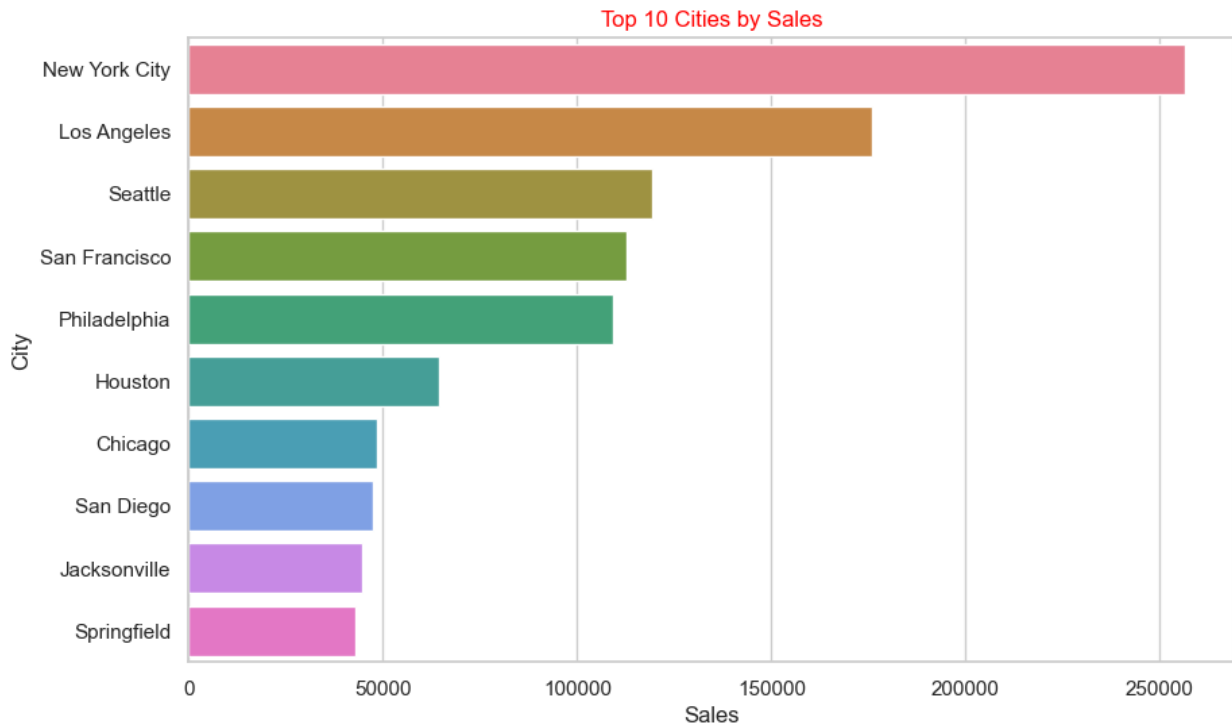
City
New York City    256319.0410
Los Angeles      175831.9010
Seattle          119460.2820
San Francisco    112577.1720
Philadelphia      109061.4610
Houston          64441.2564
Chicago          48535.9770
San Diego        47521.0290
Jacksonville     44713.1830
Springfield      43054.3420
Name: Sales, dtype: float64

plt.figure(figsize=(10, 6))

sns.barplot(
    x=top_cities.values,
    y=top_cities.index,
    palette="husl",
    hue=top_cities.index,
    legend=False
)

plt.title("Top 10 Cities by Sales", color="red")
plt.xlabel("Sales")
plt.ylabel("City")

plt.show()
```



Observation :- This identifies the cities contributing the highest sales and helps locate strong geographical markets.

Top 10 States by Sales

```
top_states = (
    df_clean.groupby("State")["Sales"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
```

top_states

State	Sales
California	457576.2715
New York	310827.1510
Texas	170124.5418
Washington	138560.8100
Pennsylvania	116496.3620
Florida	89473.7080
Illinois	80162.5370
Ohio	77976.7640
Michigan	75879.6440
Virginia	70636.7200

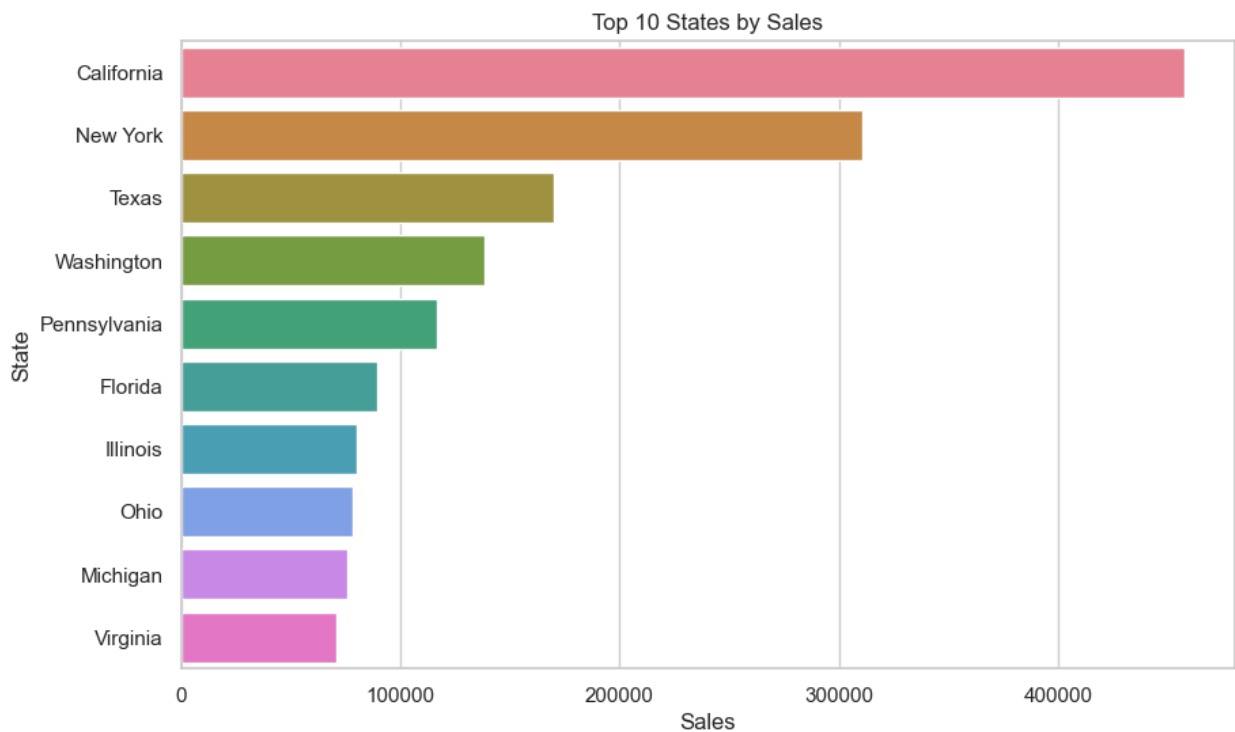
Name: Sales, dtype: float64

```
plt.figure(figsize=(10, 6))

sns.barplot(
    x=top_states.values,
    y=top_states.index,
    palette="husl"
)

plt.title("Top 10 States by Sales")
plt.xlabel("Sales")
plt.ylabel("State")

plt.show()
```



Top 10 States by Profit

```
top_profit_states = (
    df_clean.groupby("State")["Profit"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

top_profit_states
```

```
State
California    76330.7891
New York      74015.4622
Washington    33368.2375
Michigan       24428.0903
Virginia       18597.9504
Indiana        18382.9363
Georgia        16250.0433
Kentucky       11199.6966
Minnesota     10823.1874
Delaware       9977.3748
Name: Profit, dtype: float64
```

-----Quantity Analysis-----

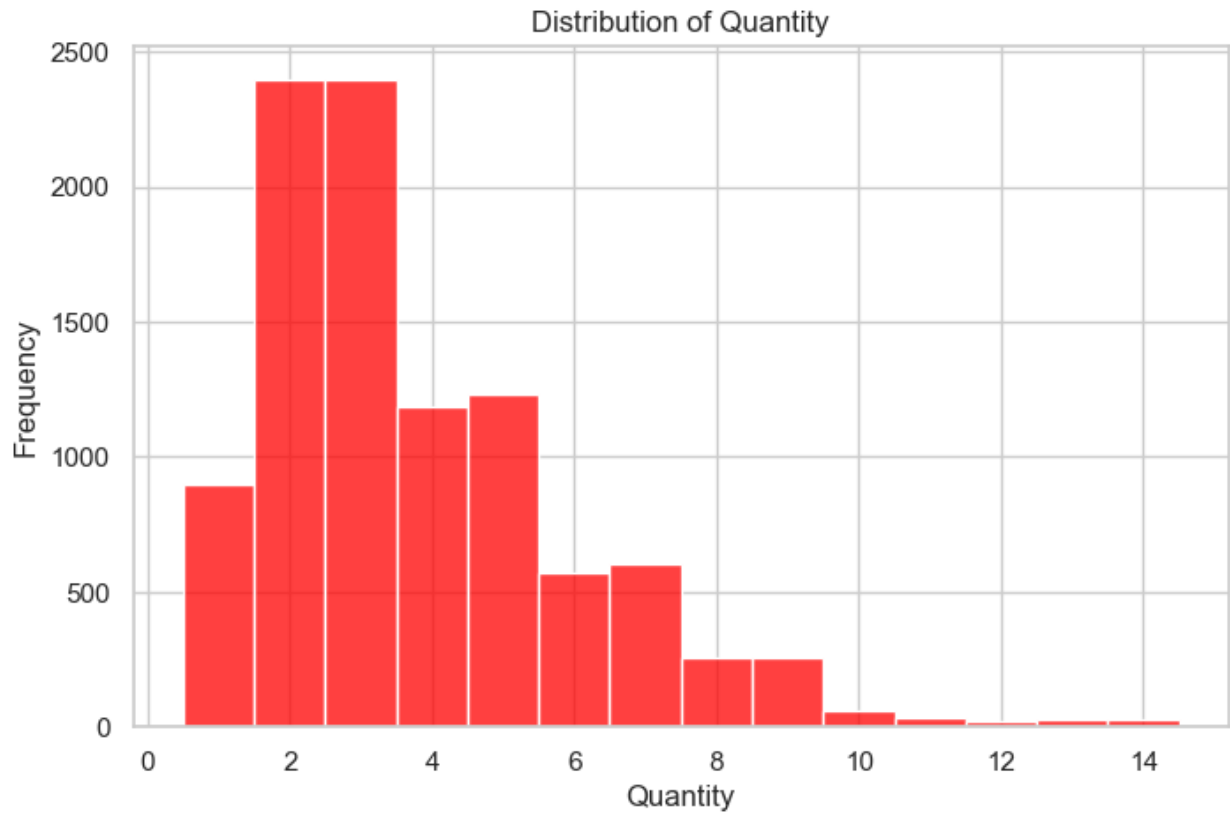
Quantity Analysis

```
plt.figure(figsize=(8, 5))

sns.histplot(
    df_clean["Quantity"],
    bins=14,
    discrete=True,
    color="red"
)

plt.title("Distribution of Quantity")
plt.xlabel("Quantity")
plt.ylabel("Frequency")

plt.show()
```



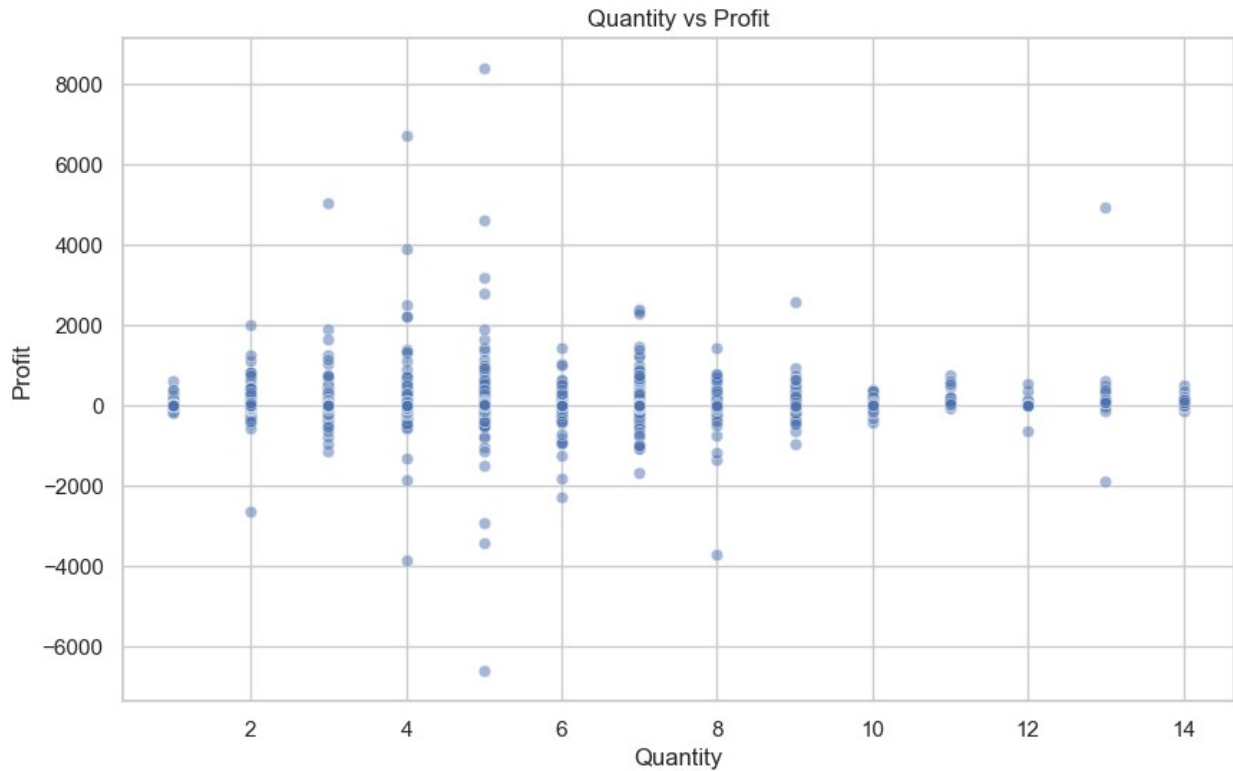
Quantity vs Profit

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df_clean,
    x="Quantity",
    y="Profit",
    alpha=0.5
)

plt.title("Quantity vs Profit")
plt.xlabel("Quantity")
plt.ylabel("Profit")

plt.show()
```



Observation :- Quantity has only a very weak positive correlation with Profit, approximately 0.07. Therefore, simply selling more units does not necessarily lead to substantially higher profit per transaction.

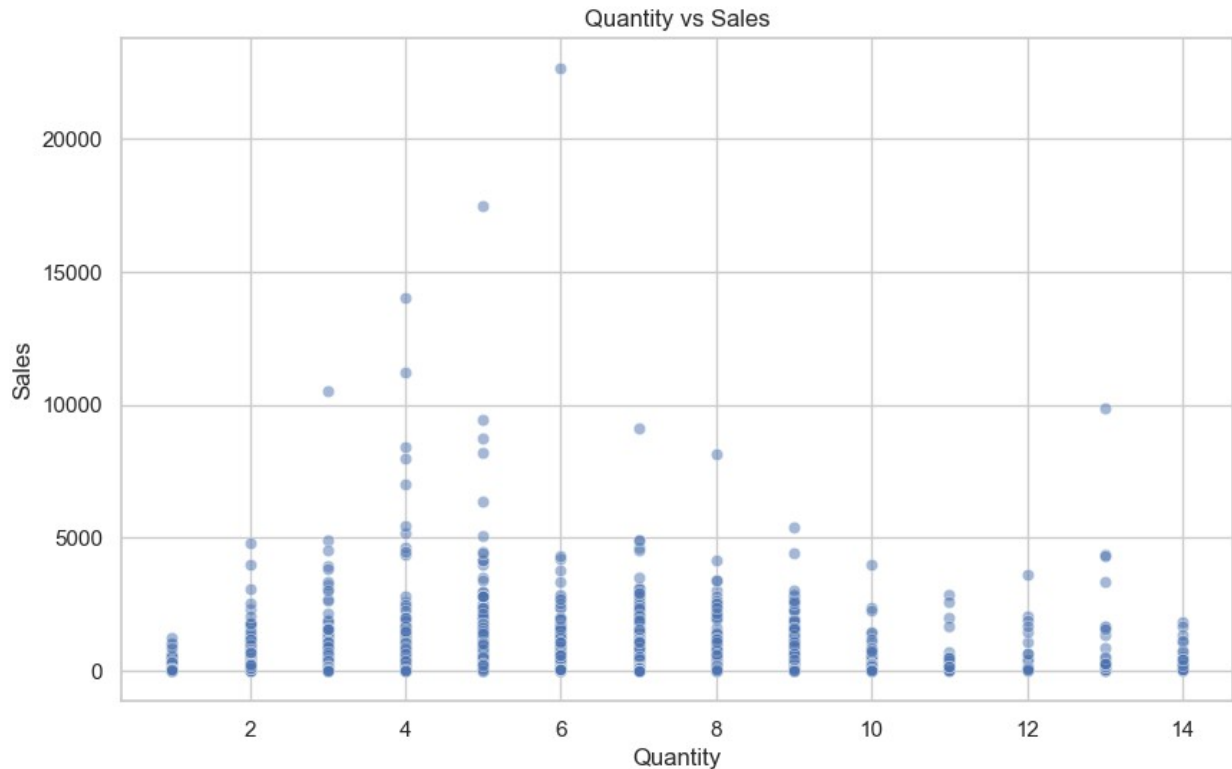
Sales vs Quantity

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df_clean,
    x="Quantity",
    y="Sales",
    alpha=0.5
)

plt.title("Quantity vs Sales")
plt.xlabel("Quantity")
plt.ylabel("Sales")

plt.show()
```



Profit Margin by Category

```
category_analysis = (
    df_clean.groupby("Category")
    .agg(
        Sales=("Sales", "sum"),
        Profit=("Profit", "sum")
    )
)

category_analysis["Profit_Margin_%"] = (
    category_analysis["Profit"] /
    category_analysis["Sales"] * 100
)
```

category_analysis

	Sales	Profit	Profit_Margin_%
Category			
Furniture	741306.3133	18421.8137	2.485047
Office Supplies	718735.2440	122364.6608	17.024998
Technology	836154.0330	145454.9481	17.395712

Key observation :- Furniture is the major concern. It generates approximately \$742K in sales, but only about \$18.5K profit, giving a margin of approximately 2.5%.

Profit Margin by Region

```
region_analysis = (  
    df_clean.groupby("Region")  
    .agg(  
        Sales=("Sales", "sum"),  
        Profit=("Profit", "sum")  
    )  
)  
  
region_analysis["Profit_Margin_%"] = (  
    region_analysis["Profit"] /  
    region_analysis["Sales"] * 100  
)  
  
region_analysis.sort_values(  
    "Profit_Margin_%",  
    ascending=False  
)
```

	Sales	Profit	Profit_Margin_%
Region			
West	725255.6365	108329.8079	14.936776
East	678435.1960	91506.3092	13.487848
South	391721.9050	46749.4303	11.934342
Central	500782.8528	39655.8752	7.918777

-----Analyze Negative Profit by Discount-----

Analyze Negative Profit by Discount

```
negative_profit_by_discount = (  
    df_clean[df_clean["Profit"] < 0]  
    .groupby("Discount")  
    .size()  
    .sort_values(ascending=False)  
)
```

negative_profit_by_discount

Discount	
0.20	502
0.70	418
0.80	299
0.30	207
0.40	180

```
0.60    138
0.50     66
0.32     27
0.15     17
0.45     11
0.10      4
dtype: int64
```

Find Maximum Sales Transaction

```
df_clean.loc[df_clean["Sales"].idxmax()]
```

```
Ship Mode      Standard Class
Segment        Home Office
Country        United States
City           Jacksonville
State          Florida
Postal Code     32216
Region         South
Category       Technology
Sub-Category   Machines
Sales          22638.48
Quantity        6
Discount        0.5
Profit         -1811.0784
Name: 2697, dtype: object
```

Find Maximum Profit Transaction

```
df_clean.loc[df_clean["Profit"].idxmax()]
```

```
Ship Mode      Standard Class
Segment        Corporate
Country        United States
City           Lafayette
State          Indiana
Postal Code     47905
Region         Central
Category       Technology
Sub-Category   Copiers
Sales          17499.95
Quantity        5
Discount        0.0
Profit          8399.976
Name: 6826, dtype: object
```

Find Maximum Loss Transaction

```
df_clean.loc[df_clean["Profit"].idxmin()]
```

Ship Mode	Standard Class
Segment	Consumer
Country	United States
City	Lancaster
State	Ohio
Postal Code	43130
Region	East
Category	Technology
Sub-Category	Machines
Sales	4499.985
Quantity	5
Discount	0.7
Profit	-6599.978
Name: 7772, dtype: object	

□ Top 3 Insights

1 Technology is the Strongest Overall Category

Technology generates approximately **\$836K in sales** and **\$145K in profit**, making it the strongest category in terms of both **sales and profitability**.

2 Discounting is a Major Profitability Concern

The analysis shows a **negative relationship between discount and profit**. At higher discount levels, profitability tends to decline and can become negative.

□ **Business Recommendation:** The company should review high-discount transactions and establish **better discount limits** to protect profit margins.

3 Furniture and Tables Require Attention

Furniture generates approximately **\$742K in sales** but only **\$18K in profit**, resulting in a very low profit margin of approximately **2.5%**.

The **Tables** sub-category generates more than **\$200K in sales** but records an approximate **\$17.7K loss**.

□ **Business Recommendation:** Management should investigate **pricing, discounts, product costs, and sales strategies** for the Furniture category, with particular attention to **Tables**.

□ Conclusion — Exploratory Data Analysis

□ 1. Dataset Quality

The dataset contains **9,994 records and 13 columns**. No missing values were found. **17 duplicate records** were identified, and after removing them, **9,977 unique records** remained. The cleaned dataset was suitable for further analysis.

□ 2. Statistical Analysis

Descriptive statistics were used to analyze **Sales, Quantity, Discount, and Profit**. Sales and Profit were right-skewed because of a small number of high-value transactions. Quantity was mainly concentrated around lower values. Outlier analysis identified unusually high Sales and Profit values. These outliers should be investigated rather than automatically removed because they may represent genuine business transactions.

□ 3. Category Analysis

Technology generated the highest Sales and Profit. **Office Supplies** also showed strong overall profitability. **Furniture** generated considerable Sales but comparatively low Profit. This demonstrates that high Sales do not always result in high Profit.

□ 4. Sub-Category Analysis

Copiers were among the most profitable sub-categories. **Phones, Accessories, Paper, and Binders** also contributed strongly to overall profitability. **Tables** generated the largest loss among the major sub-categories. **Bookcases and Supplies** also showed profitability concerns. Loss-making products require further analysis of pricing, discounts, and cost structures.

□ 5. Regional Analysis

The **West** region recorded the highest Sales and Profit. The **East** region also performed strongly. The **Central** region showed comparatively lower profitability. The **South** region generated lower Sales but maintained a reasonable contribution to overall Profit. Regional strategies should therefore be adjusted according to profitability.

□ 6. Customer Segment Analysis

The **Consumer** segment generated the highest overall Sales. **Corporate** and **Home Office** also contributed significantly. Segment analysis helps identify the most valuable customer groups.

□ 7. Shipping Analysis

Standard Class was the most frequently used shipping method. **Second Class** and **First Class** were also commonly used. **Same Day** shipping had comparatively fewer transactions. Shipping patterns can help optimize delivery services and operational costs.

□ 8. Discount & Profit Analysis

Discount showed a **negative relationship with Profit**. Higher discounts were generally associated with lower profitability. Some high-discount transactions resulted in negative Profit. Excessive discounting can therefore reduce business margins, so discount policies should be carefully monitored.

□ 9. Correlation Analysis

Sales and Profit showed a moderate positive relationship. **Sales and Quantity** showed a weak positive relationship. **Discount and Profit** showed a negative relationship. **Quantity and Profit** showed only a weak relationship. Correlation indicates relationships between variables but does not prove causation.

□ 10. Visualization Analysis

Histograms were used to analyze numerical distributions. Boxplots were used to identify potential outliers. Bar charts were used to compare Categories, Sub-Categories, Regions, Segments, and Ship Modes. Scatter plots were used to analyze relationships between Sales, Profit, Quantity, and Discount. A correlation heatmap provided an overall view of numerical relationships.

□ 11. Top 3 Key Insights

□ Technology is the Strongest Major Category

Technology is the strongest major category in terms of both **Sales and Profit**.

□ Higher Discounts Reduce Profitability

Higher discounts are associated with **lower profitability**, indicating that excessive discounting can create losses.

□ High Sales Do Not Always Guarantee High Profit

High Sales do not always result in high Profit, as demonstrated by **Furniture** and loss-making sub-categories such as **Tables**.

□ 12. Business Recommendations

- Focus on highly profitable **Categories and Sub-Categories**.
 - Review pricing and discount strategies for loss-making products.
 - Investigate the reasons behind low profitability in **Furniture**, especially **Tables**.
 - Monitor high-discount transactions carefully.
 - Analyze regional performance to improve low-margin regions.
 - Evaluate business performance using both **Sales and Profit Margin**.
 - Investigate extreme outliers for potential data-quality issues.
-

□ Final Conclusion

Overall, the EDA provided a clear understanding of the Superstore dataset and revealed important patterns in **Sales, Profit, Discount, Quantity, Categories, Sub-Categories, Regions, and Customer Segments**.

The analysis demonstrates that business performance should not be evaluated using **Sales alone**. **Profitability, discount levels, product performance, and regional trends** are equally important.

These findings can support better **pricing, discount management, product selection, and regional business strategies**.

EDA therefore provides a strong foundation for transforming raw data into meaningful insights and supporting **data-driven business decisions**.

□ Interview Questions — Exploratory Data Analysis

1. How do you distinguish a genuine outlier from a data entry error?

Answer:

I distinguish a genuine outlier from a data entry error by checking the **business context, source data, and related columns**.

- First, I identify the outlier using methods such as **IQR or Z-score**.
- I check whether the value is logically possible.
- I compare it with related columns such as **Quantity, Sales, Discount, and Profit**.
- I check the original/source data to confirm whether the value was entered correctly.
- If the value represents a real business transaction, I **keep it**.
- If it is caused by a typo, incorrect unit, or invalid entry, I **correct or remove it** based on the situation.

Key Point: I never remove an outlier automatically just because it is statistically unusual.

2. What's the difference between correlation and causation, and why does it matter in EDA?

Answer:

Correlation means that two variables are statistically related, while **causation** means that a change in one variable directly causes a change in another variable.

Example:

In this dataset:

- **Discount and Profit** have a negative correlation.
- This means higher discounts are generally associated with lower Profit.
- However, correlation alone does **not prove** that Discount is the only reason for the decrease in Profit.
- Other factors such as **product cost, pricing, category, and region** may also affect Profit.

Why does it matter?

It matters because confusing **correlation with causation** can lead to incorrect conclusions and poor business decisions.

Key Point:

Correlation shows a relationship, but it does not necessarily mean one variable causes the other.

3. Which chart type would you use to show a trend over 12 months, and why?

Answer:

I would use a **Line Chart** to show a trend over 12 months.

Reasons:

- A line chart is designed to show **changes over time**.
- It clearly shows **increases, decreases, peaks, and fluctuations**.
- Months can be arranged chronologically on the X-axis.
- It makes it easier to identify **seasonal patterns and trends**.
- It allows us to compare performance across different months.

Example:

I could use a line chart to display **monthly Sales or monthly Profit** and identify which months performed better or worse.

Key Point:

A line chart is the best choice for showing trends and changes over a continuous time period.

□ Exploratory Data Analysis — Sample Superstore

Project By: Ratnesh Chauhan

Task: Task 2 — Exploratory Data Analysis (EDA) **Level:** Level 1 **Day:** Day 2 **Tools:** Python, Pandas, Matplotlib, Seaborn

Author

Ratnesh Chauhan

This project was completed as part of the **Data Analytics Track** to analyze the Sample Superstore dataset, identify important patterns, relationships, outliers, and trends, and derive meaningful business insights through statistical analysis and data visualization.

